Physics 9702 Paper 4 · Topic 16

Thermodynamics

The seven hardest A Level Structured Questions on this topic from the 2020 to 2026 papers, hardest first, followed by their mark schemes.

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1 9702/42 February/March 2024, Question 2 (11 marks) page 2

Examiner report: (a) Many responses were seen that gave the textbook definition of the internal energy of a system in general, but only the strongest candidates then applied this correctly to the ideal gas system that was the subject of the question. Otherwise there were few difficulties, and many candidates obtained the correct final answer.

2 9702/42 May/June 2023, Question 3 (9 marks) page 4

Examiner report: (ii) Candidates found this question difficult. Only the strongest candidates appreciated that the work done by the gas in stage CD is greater than the work done on the gas in stage AB and that, over a complete cycle, the gas therefore does work that must come from a net input of thermal energy.

3 9702/41 May/June 2023, Question 4 (12 marks) page 6

Examiner report: (iii) Only the strongest candidates correctly completed this table in full. Many candidates found it difficult to correctly apply the first law of thermodynamics for the three changes.

4 9702/42 October/November 2025, Question 4 (10 marks) page 8

Examiner report: (b) (i) Only the strongest candidates were able to conclude, when comparing Fig. 4.1 and Fig. 4.2, that the internal energy at each of the stages A, B, C and D is given by U = (3/2)pV. Again, there were some difficulties over the sign of the answer.

5 9702/41 May/June 2025, Question 4 (8 marks) page 10

Examiner report: (b) Only the strongest candidates obtained full credit for this question, but many candidates earned partial credit for the work done and the increase in internal energy during cooling.

6 9702/41 October/November 2023, Question 2 (12 marks) page 11

Examiner report: Generally only the strongest candidates recognised that there was the same gain in internal energy and so the same temperature rise.

7 9702/41 October/November 2020, Question 2 (9 marks) page 13

Examiner report: (c) Completing this table proved to be challenging. Candidates had difficulty in relating the numbers they already knew to this table for the whole cycle of changes.

Questions were chosen using the Cambridge examiner reports (Principal Examiner Report for Teachers): parts the reports describe as challenging, poorly answered, or answered fully only by the strongest candidates. May/June 2020 and May/June 2026 have no report, so questions from those series were judged directly. Variant 43 repeats variant 41 in every series, so only 41 is cited.

2 **(a)** By referring to both kinetic energy and potential energy, explain what is meant by the internal energy of an ideal gas.

.....

.....

.....

..... [2]

(b) A fixed mass of an ideal gas at a temperature of 20 °C is sealed in a cylinder by a piston, as shown in Fig. 2.1.

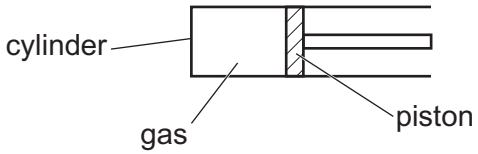


Fig. 2.1

The initial volume of the gas is $1.24 \times 10^{-4}\,m^3$.
Thermal energy is supplied to the gas and its volume increases by $5.20 \times 10^{-5}\,m^3$.

(i) The piston is freely moving so that the gas is always at atmospheric pressure.

Atmospheric pressure is $1.01 \times 10^5\,Pa$.

Calculate the work done by the gas.

work done by gas = ... J [2]

(ii) Calculate the final thermodynamic temperature T of the gas.

T = ... K [2]

(iii) The mass of the gas is 16 g. For this expansion, there is a net transfer of 960 J of thermal energy to the gas.

Calculate the specific heat capacity c of the gas at this pressure.

c = ... $J\,kg^{-1}\,K^{-1}$ [2]

(c) The gas in **(b)** is allowed to return to its starting temperature. The piston is now fixed in position.

Thermal energy is supplied to increase the temperature to the same final temperature as in **(b)**.

Use the first law of thermodynamics to suggest and explain how the specific heat capacity of the gas for this situation compares with the value in **(b)(iii)**.

.....

.....

.....

..... [3]

[Total: 11]

3 **(a)** State the first law of thermodynamics. Identify the meaning of any symbols that you use.

.....

.....

..... [2]

(b) The state of an ideal gas is continuously changed according to the cycle ABCDA shown in Fig. 3.1.

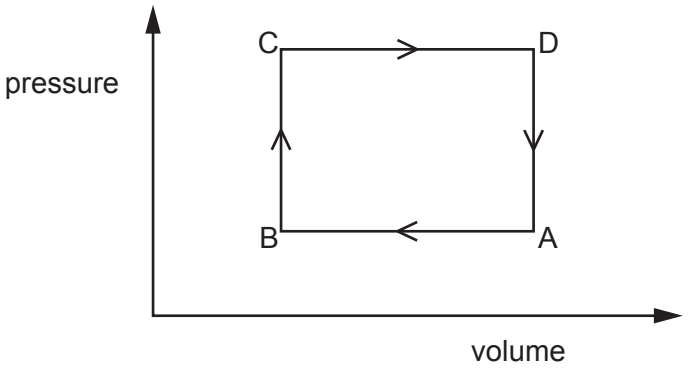


Fig. 3.1

(i) Complete Table 3.1 for the changes A to B and B to C by placing two ticks (✓) in each row.

Table 3.1

change	change in internal energy			work done on gas		
	decrease	no change	increase	negative	zero	positive
A to B						
B to C						

[4]

(ii) Use the first law of thermodynamics to describe and explain the energy transfers associated with one complete cycle ABCDA.

.....

.....

.....

.....

..... [3]

[Total: 9]

4 **(a)** State **two** of the basic assumptions of the kinetic theory of gases.

1 .....

.....

2 .....

.....

[2]

(b) An ideal gas has amount of substance n.
The gas is initially in state X, with pressure $2p$ and volume V.
The gas is cooled at constant volume to state Y, with pressure p.
The gas is then heated at constant pressure to state Z, with volume $2V$.
Finally, the gas returns at constant temperature to state X.

(i) Determine an expression for the temperature T of the gas in state X, in terms of n, p and V.
Identify any other symbols that you use.

[2]

(ii) On Fig. 4.1, sketch the variation with volume of pressure for the gas as the gas undergoes the three changes. The state X is labelled. Label states Y and Z.

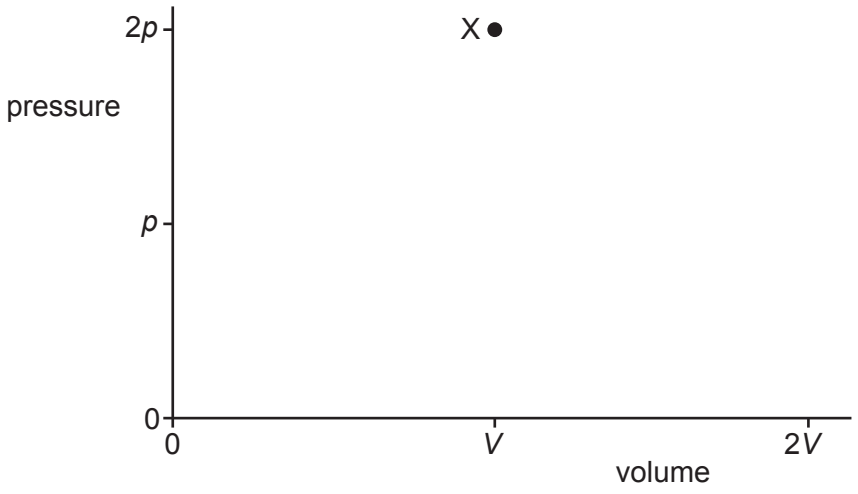


Fig. 4.1

[3]

(iii) During the change of state from Y to Z, the increase in internal energy of the gas is U. During the change of state from Z to X, the work done on the gas is W.

Complete Table 4.1 to indicate, for each of the three changes of state, the increase in internal energy of the gas, the thermal energy transferred to the gas and the work done on the gas, in terms of p, V, U and W.

Table 4.1

change	increase in internal energy of gas	thermal energy transferred to gas	work done on gas
X to Y			
Y to Z	$+U$		
Z to X			$+W$

[5]

[Total: 12]

4 A cylinder contains a fixed mass of an ideal gas at pressure 2*Y* and volume 6*X*.

The gas undergoes a sequence of changes from its initial state A, through states B, C and D, then finally back to its initial state A, as shown in Fig. 4.1.

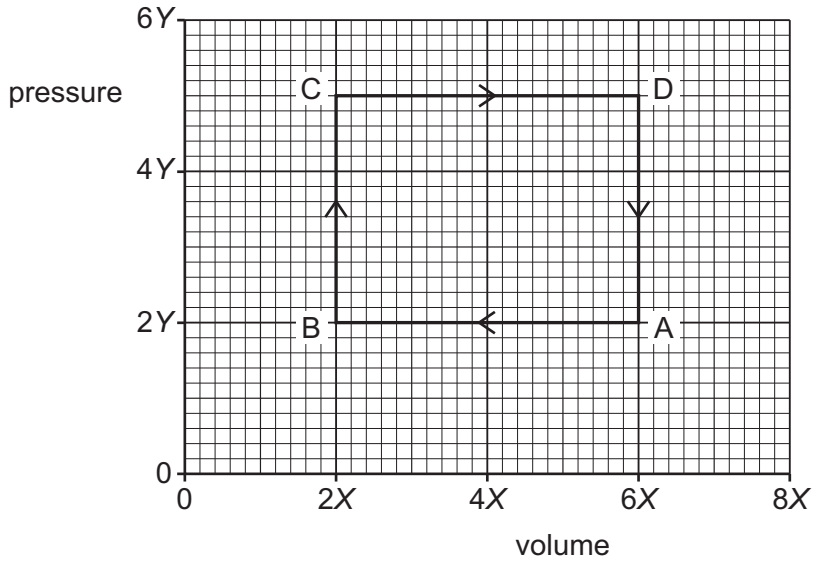


Fig. 4.1

Fig. 4.2 shows the variation with time of the internal energy of the gas.

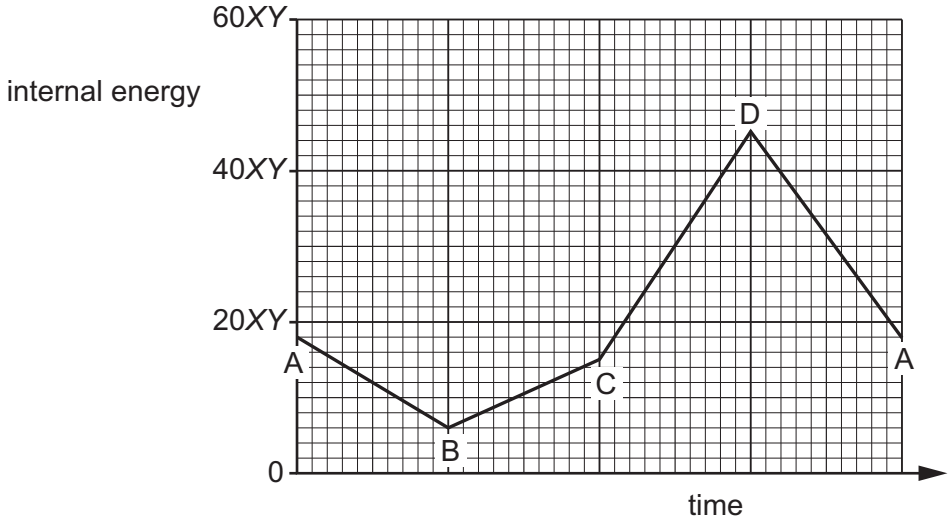


Fig. 4.2

(a) State the first law of thermodynamics.

.....

.....

..... [2]

(b) **(i)** Use Fig. 4.1 and Fig. 4.2 to determine the general expression for the internal energy U of the gas when it has pressure p and volume V.

U = ... [1]

(ii) An ideal gas at thermodynamic temperature T contains N molecules.

Use your answer in **(b)(i)** and the equation of state for an ideal gas to deduce an expression for U in terms of N and T. Identify any other symbols you use.

U = ... [2]

(c) Determine expressions, in terms of X and Y, for the work W done on the gas during:

(i) change AB

W = ... [1]

(ii) change CD.

W = ... [1]

(d) Use your answers in **(c)** and the first law of thermodynamics to determine an expression, in terms of X and Y, for the net thermal energy Q supplied to the gas during one full cycle ABCDA. Explain your reasoning.

Q = ... [3]

[Total: 10]

4 **(a)** **(i)** State what is meant by the internal energy of a system.

.....

.....

..... [2]

(ii) Explain why the internal energy of an ideal gas is directly proportional to the thermodynamic temperature of the gas.

.....

.....

.....

..... [2]

(b) A sample of an ideal gas at thermodynamic temperature T has internal energy U.

The gas is compressed so that its temperature increases to $3T$.
During this compression, work W is done on the gas.

The gas is then cooled at constant volume so that its temperature decreases to $2T$.

Complete Table 4.1 to show, in terms of some or all of W, T and U, the work done on the gas, the thermal energy supplied to the gas and the increase in internal energy of the gas for each of the two processes.

Table 4.1

	work done on gas	thermal energy supplied to gas	increase in internal energy of gas
compression	$+W$		
cooling			

[4]

[Total: 8]

2 **(a)** Define specific heat capacity.

.....

.....

..... [2]

(b) An ideal gas of mass 0.35 kg is heated at a constant pressure of 2.0×10^5 Pa so that its internal energy increases by 7600 J. During this process, the volume of the gas increases from 0.038 m^3 to 0.063 m^3 and the temperature increases by 56 °C.

(i) Show that the magnitude of the work done on the gas is 5000 J.

[1]

(ii) Explain whether the work done on the gas is positive or negative.

.....

.....

..... [2]

(iii) Determine the magnitude of the thermal energy q transferred to the gas.

q = ... J [2]

(iv) Calculate the specific heat capacity of the gas for this process. Give a unit with your answer.

specific heat capacity = ... unit ..... [2]

(c) The gas in **(b)** is now heated at constant volume rather than at constant pressure. The increase in internal energy of the gas is the same as in **(b)**.

Use the first law of thermodynamics to explain whether the specific heat capacity of the gas for this process is less than, the same as, or greater than the answer in **(b)(iv)**.

.....

.....

.....

.....

..... [3]

[Total: 12]

2 **(a)** The first law of thermodynamics may be expressed as

$$\Delta U = (+q) + (+w)$$

where ΔU is the increase in internal energy of the system.

State the meaning of:

$+q$ .....

.....

$+w$. .....

.....

[2]

(b) The variation with pressure p of the volume V of a fixed mass of an ideal gas is shown in Fig. 2.1.

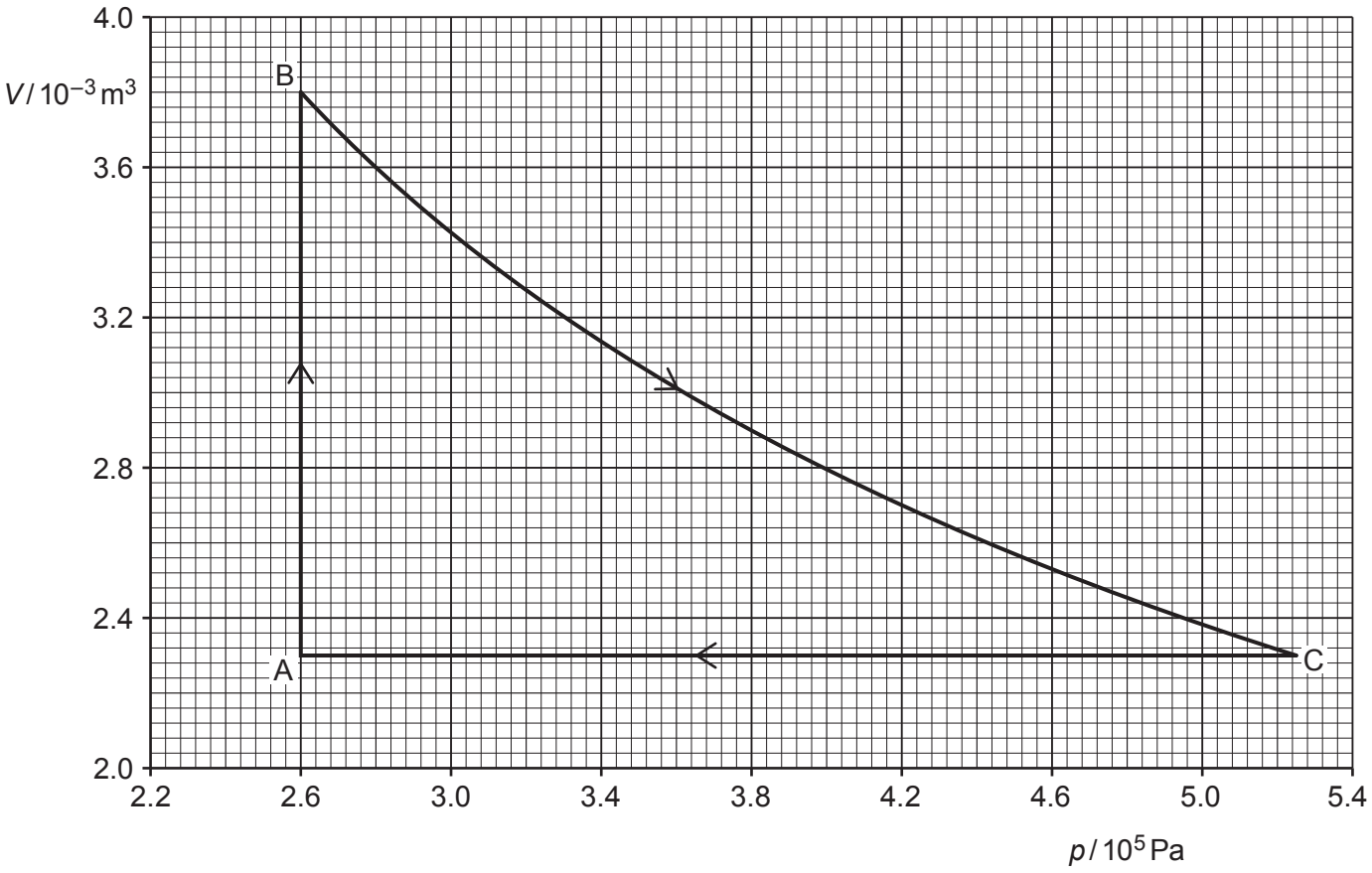


Fig. 2.1

The gas undergoes a cycle of changes A to B to C to A.

During the change A to B, the volume of the gas increases from $2.3 \times 10^{-3}\,m^3$ to $3.8 \times 10^{-3}\,m^3$.

(i) Show that the magnitude of the work done during the change A to B is 390 J.

[1]

(ii) State and explain the total change, if any, in the internal energy of the gas during one complete cycle.

.....

.....

..... [2]

(c) During the change A to B, 1370 J of thermal energy is transferred to the gas.

During the change B to C, no thermal energy enters or leaves the gas. The work done on the gas during this change is 550 J.

Use these data and the information in **(b)** to complete Table 2.1.

Table 2.1

change	q / J	w / J	ΔU / J
A to B	.....	.....	.....
B to C	.....	.....	.....
C to A	.....	.....	.....

[4]

[Total: 9]

Question	Answer	Marks
2(a)	total kinetic energy associated with random motion of molecules	**M1**
	plus total potential energy (of molecules) but potential energy is zero	**A1**
2(b)(i)	$W = p\Delta V$	**C1**
	$= 1.01 \times 10^5 \times 5.20 \times 10^{-5}$ $= (+)5.25$ J	**A1**
2(b)(ii)	$V \propto T$ *or* V / T = constant	**C1**
	$1.24 / (273 + 20) = (1.24 + 0.520) / T$ $T = 416$ K	**A1**
2(b)(iii)	$c = Q / m\Delta T$	**C1**
	$= 960 / (0.016 \times (416 - 293))$ $= 490$ J kg^{-1} K^{-1}	**A1**
2(c)	no change in volume so no work is done (by the gas)	**B1**
	(same temperature change so) same change in internal energy	**B1**
	less thermal energy needs to be supplied so c is less	**B1**

Question	Answer	Marks
3(a)	change in internal energy = work done + energy transfer by heating	C1
	increase in internal energy = work done on system + energy transferred to the system by heating	A1
3(b)(i)	AB change in internal energy: decrease	B1
	AB work done on gas: positive	B1
	BC change in internal energy: increase	B1
	BC work done on gas: zero	B1
3(b)(ii)	more work done by gas in CD than is done on gas in AB **or** (no work done on gas in BC and DA so) (overall) gas does work	B1
	(overall) change in internal energy is zero	B1
	(must be an overall) input of thermal energy	B1

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>4(a)</td><td>• particles are in (continuous) random motion
• particles have negligible volume (compared with the gas)
• negligible forces between particles (except during collisions)
• (all) collisions (perfectly) elastic
• time of collision negligible (in comparison with time between collisions)
Any two points, 1 mark each</td><td>B2</td></tr>
<tr><td rowspan="2">4(b)(i)</td><td>(general starting equation) $pV = nRT$</td><td>C1</td></tr>
<tr><td>$T = (2pV / nR)$ where R is the (molar) gas constant</td><td>A1</td></tr>
<tr><td rowspan="3">4(b)(ii)</td><td>sketch: straight vertical line XY from (V, $2p$) to (V, p)</td><td>B1</td></tr>
<tr><td>straight horizontal line YZ from (V, p) to ($2V$, p)</td><td>B1</td></tr>
<tr><td>curve with gradient increasing from Z to X from ($2V$, p) to (V, $2p$)</td><td>B1</td></tr>
<tr><td rowspan="5">4(b)(iii)</td><td>XY work done on gas correct (= 0)</td><td>B1</td></tr>
<tr><td>ZX increase in internal energy correct (= 0)</td><td>B1</td></tr>
<tr><td>YZ work done on gas correct (= $–pV$)</td><td>B1</td></tr>
<tr><td>XY increase in internal energy such that the increase in internal energy column adds up to zero</td><td>B1</td></tr>
<tr><td>all three thermal energies transferred such that $\Delta U = q + w$ in each row

(completely correct answer:

<table><tr><td>change</td><td>ΔU</td><td>Q</td><td>w</td></tr><tr><td>X to Y
Y to Z
Z to X</td><td>$–U$
[$+U$]
0</td><td>$–U$
$U + pV$
$–W$</td><td>0
$–pV$
[$+W$]</td></tr></table>
)</td><td>B1</td></tr>
</table>

Question	Answer	Marks
4(a)	change in internal energy = work done + energy transfer by heating	**C1**
	increase in internal energy = work done on system + energy transferred to the system by heating	**A1**
4(b)(i)	$U = (3 / 2)\ pV$	**A1**
4(b)(ii)	$pV = NkT$ and k identified as Boltzmann constant	**B1**
	$U = (3 / 2)\ NkT$	**A1**
4(c)(i)	$W = (+)8XY$	**A1**
4(c)(ii)	$W = –20XY$	**A1**
4(d)	work done during stages BC and DA = 0	**B1**
	change in internal energy (over complete cycle) = 0	**C1**
	thermal energy supplied $= 20XY – 8XY$ $= (+)12XY$	**A1**

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>4(a)(i)</td><td>sum of potential energy and kinetic energy</td><td>B1</td></tr>
<tr><td></td><td>(total) energy of random motion of particles</td><td>B1</td></tr>
<tr><td>4(a)(ii)</td><td>potential energy (of molecules) (in an ideal gas) is zero, so the internal energy of the gas is equal to the total kinetic energy (of molecules)</td><td>B1</td></tr>
<tr><td></td><td>kinetic energy of molecules is proportional to (thermodynamic) temperature (so internal energy is proportional to (thermodynamic) temperature))</td><td>B1</td></tr>
<tr><td>4(b)</td><td>cooling work done = 0</td><td>B1</td></tr>
<tr><td></td><td>compression increase in internal energy = +2U</td><td>B1</td></tr>
<tr><td></td><td>cooling change in internal energy = –U</td><td>B1</td></tr>
<tr><td></td><td>both rows: thermal energy adds to work to give increase in internal energy in terms of U and/or W

(if fully correct, thermal energy for compression = 2U – W and thermal energy for cooling = –U:
<table><tr><td>compression</td><td>+W</td><td>2U – W</td><td>+2U</td></tr><tr><td>cooling</td><td>0</td><td>–U</td><td>–U</td></tr></table>)</td><td>B1</td></tr>
</table>

Question	Answer	Marks
2(a)	(thermal) energy per unit mass (to change temperature)	**B1**
	(thermal) energy per unit change in temperature	**B1**
2(b)(i)	work done $= p\Delta V$ $= (2.0 \times 10^5) \times (0.063 - 0.038) = 5000\ \text{J}$	**A1**
2(b)(ii)	gas is expanding (against external pressure)	**B1**
	gas does work / work is done by gas, so (work done on gas is) negative	**B1**
2(b)(iii)	$\Delta U = q + W$	**C1**
	$7600 = q + (-5000)$ $q = 12\,600\ \text{J}$	**A1**
2(b)(iv)	specific heat capacity $= q / m\Delta T$ $= 12600 / (0.35 \times 56)$	**C1**
	$= 640\ \text{J kg}^{-1}\ \text{K}^{-1}$	**A1**
2(c)	same gain in internal energy so same temperature rise	**B1**
	no change in volume so no work done **or** no work done so less thermal energy needed (for same change in internal energy)	**B1**
	less thermal energy needed (for same temperature change) so lower specific heat capacity	**B1**

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>2(a)</td><td>+q: thermal energy transfer to system</td><td>B1</td></tr>
<tr><td></td><td>+w: work done on system</td><td>B1</td></tr>
<tr><td>2(b)(i)</td><td>(W =) $2.6 \times 10^5 \times (3.8 - 2.3) \times 10^{-3}$ = 390 J</td><td>A1</td></tr>
<tr><td>2(b)(ii)</td><td>no (total) change (in internal energy)</td><td>B1</td></tr>
<tr><td></td><td>gas returns to its original temperature</td><td>B1</td></tr>
<tr><td>2(c)</td><td>A to B row all correct (1370, – 390, 980)</td><td>B1</td></tr>
<tr><td></td><td>B to C row all correct (0, 550, 550)</td><td>B1</td></tr>
<tr><td></td><td>C to A row: ΔU adds to the other two ΔU values to give zero</td><td>B1</td></tr>
<tr><td></td><td>C to A row: w = 0 and q adds to w to give ΔU value

complete correct answer:

<table>
<tr><th>change</th><th>q / J</th><th>w / J</th><th>ΔU / J</th></tr>
<tr><td>A to B
B to C
C to A</td><td>(+)1370
0
–1530</td><td>–390
(+)550
0</td><td>(+)980
(+)550
–1530</td></tr>
</table>
</td><td>B1</td></tr>
</table>